

The empirical recurrence for OEIS A251612: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

2 September 2026

Abstract

OEIS A251612 records “Empirical recurrence of order 49”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 343$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 49, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 49 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A251612 is “Number of $(n+1) \times (2+1)$ 0..6 arrays with every 2×2 subblock summing to a prime.”. It has offset 1 and begins

14008, 583034, 24317054, 1016045742, 42444702494, 1774758313314, 74187807802266, 3103654124204860, ...

The entry states:

Empirical recurrence of order 49 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 13:28:09, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 343$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 343$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 686$ exact terms of the model returns order 49 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{49} c_i a(n-i),$$

c_1	61	c_{14}	−64374099723210953518	c_{27}	82220384849092534373583164661
c_2	2656	c_{15}	−186035369688519491044	c_{28}	122809852718951244598465222402
c_3	−216061	c_{16}	8484687238494645519719	c_{29}	−131083356288962311583670091765
c_4	−1387330	c_{17}	−332076787670414826261	c_{30}	−61850959315271519862421160666
c_5	280701862	c_{18}	−763606912279457261984135	c_{31}	1391818660446107313208307007716
c_6	−1257360157	c_{19}	1528891178835577200107819	c_{32}	−262922385193035991227200050152
c_7	−188408927989	c_{20}	47369765992817229466574374	c_{33}	−971508543040544674613369459512
c_8	1887730050237	c_{21}	−163753189358254388221865916	c_{34}	6189848263376145429582790865965
c_9	73784606748362	c_{22}	−2018099047187261369654612100	c_{35}	4324498272905082714146360496424
c_{10}	−1040441316624966	c_{23}	9445850842555674961217448440	c_{36}	−4132749910190347450525277071835
c_{11}	−17590975813624055	c_{24}	58053623136081071763010615688	c_{37}	−1158821358606709433495433349211
c_{12}	325194870904798787	c_{25}	−343613069868783372637241984000	c_{38}	13852162097904365107829833855677
c_{13}	2514506146011920374	c_{26}	−1088361018803076224269643294624	c_{39}	16868604549278320731064271841420

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 49, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $49 + 4$ terms removed returns order 49 as well, so $\deg q_{\min} = 49 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 13 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $49 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 49 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A251612.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.